Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	6 February 2026
Team ID	LTVIP2026TMIDS81059
Project Name	Visualizing Housing Market Trends: An Analysis of Sale Prices and Features using Tableau
Maximum Marks	4 Marks

Technical Architecture:

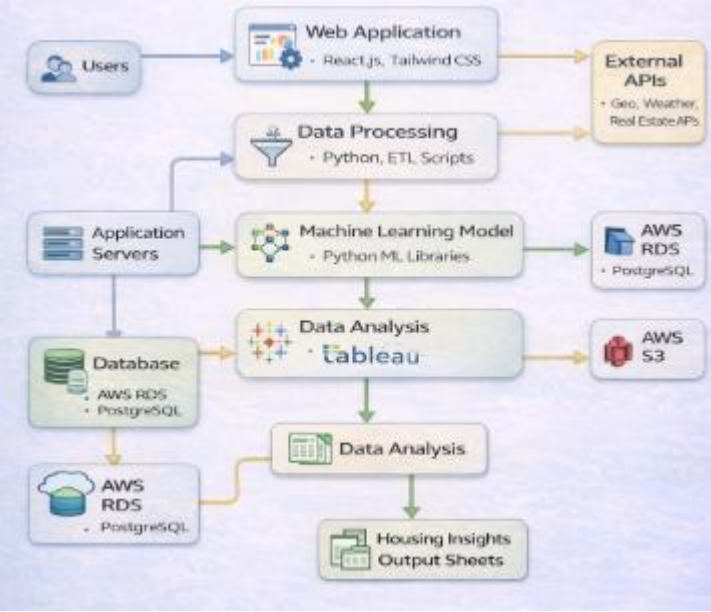

Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Interactive housing dashboard accessible via browser	Tableau Public / Tableau Desktop
2	Data Processing	Cleaning, transforming housing dataset (ETL)	Python (Pandas, NumPy)
3	Data Analytics Engine	Feature correlation, pricing trend analysis	Python, Statistical Models
4	Visualization Layer	Charts, histograms, bar graphs, story dashboards	Tableau
5	Database	Store cleaned housing dataset	PostgreSQL / MySQL
6	File Storage	Store exported reports & datasets	Local Storage / AWS S3
7	External Data (Optional)	Real estate market indicators (if integrated)	Real Estate APIs
8	Infrastructure	Hosting & deployment (if web-enabled)	AWS EC2 / Local Server
9	Reporting Module	Export pricing insights	PDF / CSV Export
10	Security Layer	Access control & data protection	HTTPS, Role-Based Access

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Tools	Data processing using open-source libraries	Python, Pandas
2	Security Implementation	Authentication & secure data handling	HTTPS, Role-Based Access
3	Scalability	Supports growing housing datasets	Cloud Storage, Optimized Queries
4	Availability	Dashboard accessible anytime	Cloud Hosting / Tableau Server
5	Performance	Fast chart rendering & filter response	Optimized Tableau Extracts
6	Reliability	Accurate and validated pricing calculations	Data Validation Scripts

References

1. https://c4model.com/
2. https://www.ibm.com/cloud/architecture
3. https://aws.amazon.com/architecture
4. https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d